



JSON and REST APIs - the best of both worlds now in SQL Server 2025

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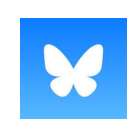
JSON and REST APIs in SQL Server 2025

Thomas Hütter, Diplom-Betriebswirt

- Developer for ERP apps, SQL scripts and BI stuff
- Worked at consultancies, ISVs, end user companies
- 1995: SQL Server, 2014: R, 2020: Power*, 2024: Arduino
- Speaker at SQL / data / dev events around Europe



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Agenda

- Introducing JSON basics ...
- ... and the JSON news in SQL Server 2025
- Introducing REST APIs ...
- ... and calling REST APIs from SQL Server
- ... Postman and Beeceptor are here to help you
- ... Demos from around the globe
- Round-up, resources, learning

Introducing JSON basics and the JSON news in SQL Server 2025



- All steps of JSON's history
- All details of JSON in char fields
- Repeat *all* JSON functions in depth



- Overview of JSON history
- Recap of JSON in char fields
- Possibilities of *new-in-SQL-2025*
JSON data type, functions, indexes

Introducing JSON basics ...

- JSON = JavaScript Object Notation („hipster’s XML“)
- In the early 2000s, a stateless, real-time server-to-browser data-interchange communication protocol was needed
- First specified by Douglas Crockford in the 2000s, lightweight alternative to XML
- consists of key-value pairs, easy to read and parse
- Specifications: RFC 8259, IETF Std 90, ECMA-404, ISO/IEC 21778:2017
- „The software shall be used for Good, not Evil“ is part of the license 😊

Introducing JSON basics ...

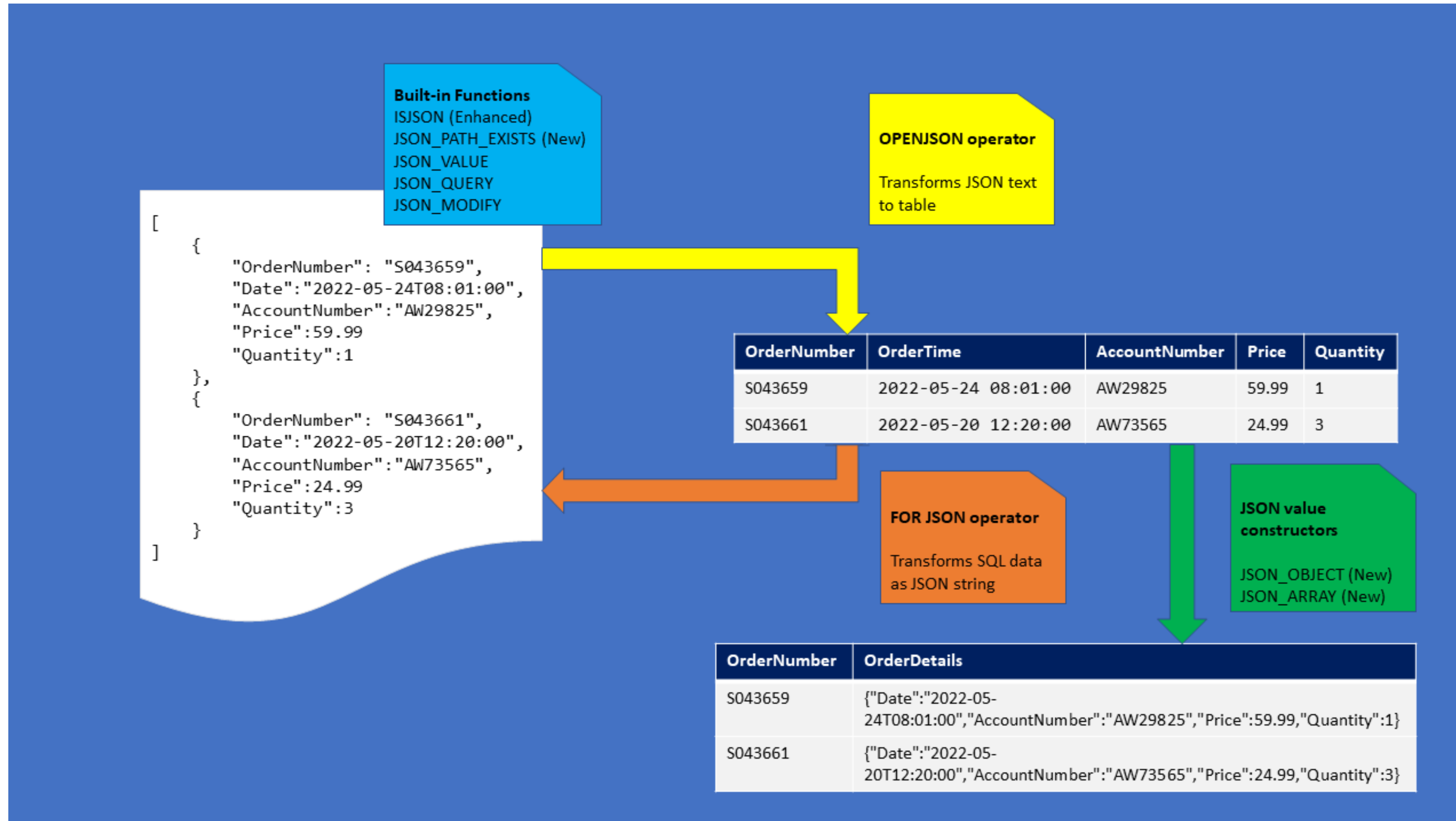
- „Why JSON? - JSON is everywhere!“ (Jovan Popovic, PM at Microsoft)
- transferring data between web servers and applications, popular for REST APIs
- IoT devices send semi-structured data in JSON format
- log and telemetry data may come JSON-formatted
- cloud services use JSON format internally (Azure Stream Analytics)
- user settings are stored in json files (e.g. VS Code and others)
- popular data format for *NoSQL* databases
- ...

Introducing JSON basics ...

- `{"event": "FabCon-SQLCon 2026", "location": "Barcelona"}`
- `{
 "event": "FabCon-SQLCon 2026",
 "location": "Barcelona"
}`
- `{
 "PersonID": 1001,
 "FullName": "Frank Geisler",
 "Languages": ["German", "English", "Catalan"]
}`

... JSON basics in SQL Server

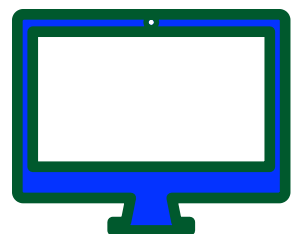
JSON support in SQL Server until 2022 based on NVarChar(4000 or max):



... and the JSON news in SQL Server 2025

SQL Server 2025 brings us:

- the native JSON data type
- and (some sort of) JSON indexes
- the JSON_CONTAINS function
- the JSON_OBJECTAGG function
- the JSON_ARRAYAGG function
- further optimization for other functions when used in conjunction with the new native JSON data type



Introducing REST APIs



Introducing REST APIs and calling REST APIs from SQL Server



- Listing all aspects of REST APIs
- How to build REST APIs
- No Swagger, no OpenAPI



- Overview of what REST APIs are
- From here on: examples
 - My use cases for tools such as your browser, Postman and Beeceptor
- How to call REST APIs from T-SQL
- How to handle the results

Introducing REST APIs

What does *REST API* stand for?

- REST(ful) = **R**epresentational **S**tate **T**ransfer

An architecture designed for loosely-coupled client-server applications.

- API = **A**pplication **P**rogramming **I**nterface

A computer-to-computer connection (as opposed to user interface),
exposing a well-defined interface while hiding the internal details.

Introducing REST APIs

What functionality does a REST API usually offer?

The possibility to retrieve, modify or even delete data in another system.

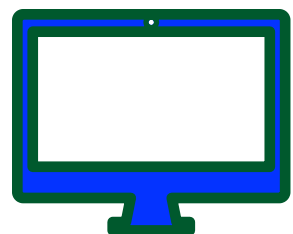
The well-known CRUD operations can be mapped to API methods:

- Create → POST: Inserts a new record (or sends otherwise needed data).
- Read → GET: Retrieves data (without modifying it).
- Update → PUT / PATCH: Replaces an existing record / modifies specific fields.
- Delete → DELETE: Removes a record from the database.

Introducing REST APIs

Some examples use cases of (publicly) available REST APIs:

- Validating data, e.g. European VAT IDs, IBANs, postal or e-mail addresses ...
- Fetching financial data - currency exchange rates, stock prices ...
- Getting general data - environmental / weather, traffic information (flights, airports), gas prices, geolocation / geocoding, Open Government ...
- Not so *public*: placing orders with your favorite online warehouse, checking shipment status, facilitating payments ...



Tools that are here to help us

Postman:

- A unified platform for designing, testing, using APIs
- Can be used in the browser or installed locally
- Collaboration with developer teams and consumers
- There is a free tier as well as Solo, Team and Enterprise plans
- Offers support, an academy and there is a community

Beeceptor:

- ‘A powerful tool for testing, mocking, and intercepting HTTP requests’
- Supports REST API, GraphQL, SOAP and gRPC endpoints
- There is a free tier as well as Individual, Team and Scale plans

...and calling REST APIs from SQL Server

Prerequisites:

- Enable calling the procedure:

```
EXECUTE sp_configure 'external rest endpoint enabled', 1;  
RECONFIGURE WITH OVERRIDE;
```

- Grant database permissions:

```
GRANT EXECUTE ANY EXTERNAL ENDPOINT TO [<PRINCIPAL>];
```

- If needed, create Master key and credentials:

```
CREATE MASTER KEY ENCRYPTION BY PASSWORD = '<YourStrongPassword!2026>';
```

```
CREATE DATABASE SCOPED CREDENTIAL [<the API URL here>]  
WITH IDENTITY = 'HTTPEndpointHeaders' | 'HTTPEndpointQueryString' | 'Managed Identity',  
SECRET = '{"Authorization": "Bearer <your bearer token>"}'
```

```
GRANT REFERENCES ON DATABASE SCOPED CREDENTIAL:: [<CRED_NAME>] TO [<PRINCIPAL>];
```

...and calling REST APIs from SQL Server

EXECUTE @returnValue = sp_invoke_external_rest_endpoint

[@url =] N'url' **nvarchar**(4000)

[, [@payload =] N'request_payload'] **nvarchar**(max)

[, [@headers =] N'http_headers_as_json_array'] **nvarchar**(4000)

[, [@method =] 'get' | '**post**' | 'put' | 'patch' | 'delete' | 'head'] **nvarchar**(6)

[, [@timeout =] seconds] **smallint**(pos.) default=30

[, [@credential =] credential] **sysname**

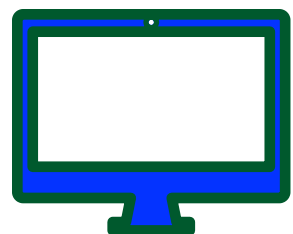
[, @response OUTPUT] **nvarchar**(max) !

[, [@retry_count =] # of retries if any errors] **tinyint**(pos.) default=0

...and calling REST APIs from SQL Server

Limitations and differences between on-prem and cloud flavours:

- Payload size < 100 MB, URL < 8 KB, headers < 8 KB
- Supported payload media types are JSON, XML, text
- `sp_invoke_external_rest_endpoint` is enabled by default in Azure SQL database and SQL databases in Fabric, needs to be enabled on-prem.
- For Azure SQL database and managed instances, there is a 'white list' of allowed endpoint domains/services - all within the Microsoft ecosystem. 😎
- Need to access any other endpoint? Take a detour via Azure API Management. 😊



Resources:

- General info on JSON: <https://www.json.org/json-en.html>, <https://en.wikipedia.org/wiki/JSON>
- MS Learn for SQL Server 2025:
on using JSON <https://learn.microsoft.com/en-us/sql/relational-databases/json/json-data-sql-server>
on calling REST APIs <https://learn.microsoft.com/en-us/sql/relational-databases/system-stored-procedures/sp-invoke-external-rest-endpoint-transact-sql>
- Tools used: <https://www.postman.com/>, <https://beeceptor.com/>
- API collections: <https://github.com/public-apis/public-apis>, <https://publicapis.dev/>, <https://free-apis.github.io/>,
<https://github.com/marcelscruz/public-apis>, <https://www.freepublicapis.com/>

JSON and REST APIs in SQL Server 2025

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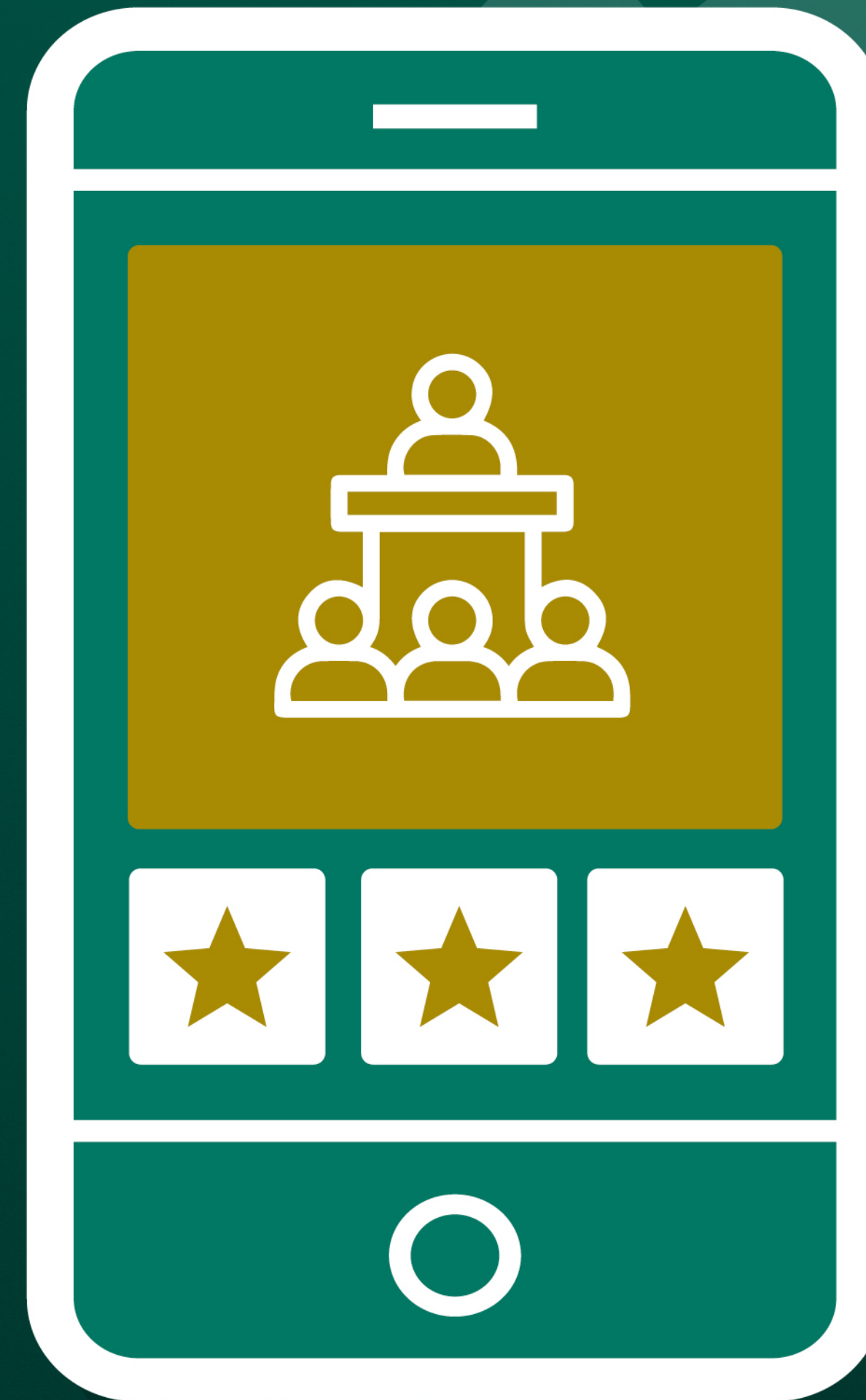
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